

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Case Study on Individual Browser History

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Course Title: Computer Forensics Case Study

Course Code: SBT-DF204

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Computer Forensics Case Studies

1. EXECUTIVE OVERVIEW

This formal technical report details the offline digital forensic examination and database analysis of the Google Chrome History SQLite database acquired from a target system associated with suspected controlled substance trafficking. The primary objective of this investigation was to reconstruct a time-ordered, evidence-based chronological account of web browser activity, examine navigation transitions, identify online advertisement postings, evaluate email communications and cryptocurrency transaction verifications, and inspect downloaded digital artifacts.

Forensic examination of the Chrome history database successfully reconstructed an interconnected workflow of suspected drug trafficking activity executed on **Tuesday, 19 April 2022** between **13:54:18 UTC and 15:02:56 UTC**. The observed activity includes:

1. **Classified Advertisement Posting:** Browsing and managing a listing for prescription supplements on Craigslist (baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html) and accessing its post management portal (post.craigslist.org/manage/7473121658).
2. **Local Evidence Retention:** Downloading and storing a proof-of-payment image file (proof_of_payment.png, 33,844 bytes) to local destination path C:\Users\FSCS_User\Desktop\proof_of_payment.png at 14:56:58 UTC.
3. **Cryptocurrency Verification & Ledger Inspection:** Inspecting a Bitcoin transaction on the Mempool.space blockchain explorer (tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5), alongside inspecting wallet funding and ledger balances ($41,563.70 USD) on the Kraken exchange (kraken.com/u/history/ledger).
4. **Email Communications:** Correlating active Gmail session threads (unsub.fscs@gmail.com) matching the listing title and Bitcoin transaction hash.

Forensic Summary & Confidence Note: Database analysis conclusively links browser user profile FSCS_User to the creation, verification, and local storage of trafficking-related artifacts. Confidence in profile-level activity is **High**. However, because browser artifacts record software profile actions rather than human identity, history evidence alone cannot rule out shared physical endpoint access, unattended active sessions, or remote desktop usage without corroborating system logon or biometric logs.

EVIDENCE INVENTORY

To maintain strict forensic chain-of-custody, cryptographic SHA-256 digests were computed immediately upon evidence acquisition and working directory initialization. All analytical procedures were performed exclusively on write-protected working duplicates within a Kali Linux forensic environment.

Evidence ID	File Path / Type	Stage SHA-256 Cryptographic Hash Digest	Description
EVID-01	evidence/History / Raw SQLite DB	b991b0fa69062c44caf599c468c8088576ebd9cebbf46d02c89cf51bcde3fd19	Primary raw Chrome History SQLite database acquired from suspect device.
EVID-02	working/History_working.sqlite / Working Copy	b991b0fa69062c44caf599c468c8088576ebd9cebbf46d02c89cf51bcde3fd19	Verified bit-stream working duplicate created for active SQLite analysis.
EVID-03	exported/proof_of_payment.png / Extracted File	N/A (33,844 bytes recorded in downloads table)	Local file artifact downloaded to C:\Users\FSCS_User\Desktop\proof_of_payment.png

2. TECHNICAL LAB EXECUTION & EVIDENCE ANALYSIS

Phase 1: Environment Initialization & Forensic Preservation

A structured directory workspace was initialized under /home/garba/Downloads/Digital_Currency/LabFiles/ on a Kali Linux virtual machine. The original file History was write-protected using file permissions (chmod 444 History), and a working duplicate History_working.sqlite was generated. Cryptographic SHA-256 digests were calculated before analysis to verify bit-stream parity.

Bash

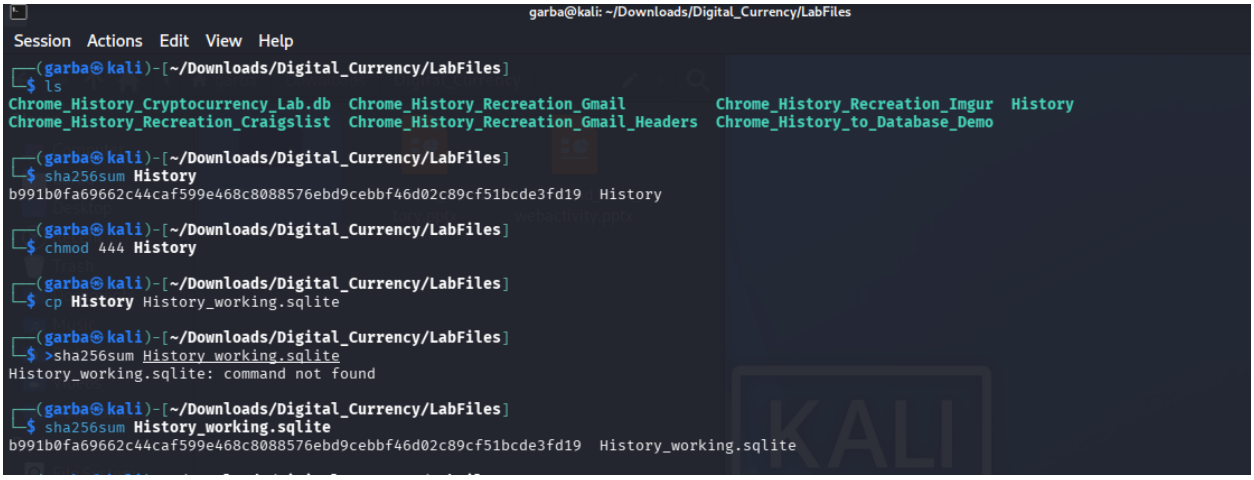
```
cd ~/Downloads/Digital_Currency/LabFiles

chmod 444 History

cp History History_working.sqlite

sha256sum History History_working.sqlite
```

Figure A1: Terminal verification of primary evidence file History and bit-stream working copy History_working.sqlite, confirming identical SHA-256 hash digests (b991b0fa69062c44caf599c468c8088576ebd9cebbf46d02c89cf51bcde3fd19). *(Reference:*



Phase 2: Database Schema & Architecture Inspection

The working database was examined using sqlite3 v3.53.4. Schema structure and field types were evaluated across four primary relational tables (urls, visits, downloads, downloads_url_chains) using PRAGMA table_info queries.

SQL

sqlite3 History_working.sqlite

.headers on

.mode column

SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;

PRAGMA table_info(urls);

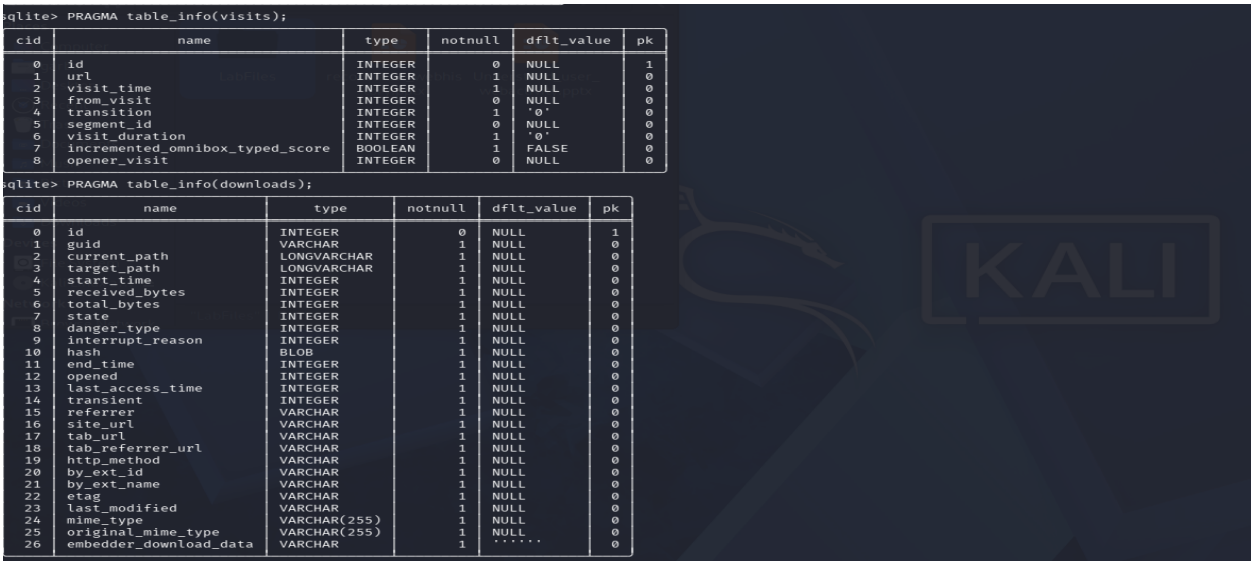
PRAGMA table_info(visits);

PRAGMA table_info(downloads);

TABLE NAME	PRIMARY / FOREIGN KEYS	CRITICAL FIELDS REVIEWED	FORENSIC FUNCTION & ANALYTICAL ROLE
urls	Primary Key: id	url, title, visit_count, typed_count	Master index storing unique web addresses and page titles.
visits	Primary Key: id Foreign Key: url → urls.id	visit_time, from_visit, transition	Individual event log recording WebKit timestamps, referral IDs, and transition bitmasks.
downloads	Primary Key: id	target_path, start_time, received_bytes, state	Host filesystem log tracking downloaded files, original desktop paths, and file sizes.
downloads_url_chains	Primary Key: id	chain_index, url	Relational table linking download IDs to source referrers and download URLs.

Figure A2: Terminal output displaying SQLite database schema inspection for urls, visits, and downloads tables.





```
sqlite> PRAGMA table_info(visits);
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	url	INTEGER	1	NULL	0
2	visit_time	INTEGER	1	NULL	0
3	from_visit	INTEGER	0	NULL	0
4	transition	INTEGER	1	'0'	0
5	segment_id	INTEGER	0	NULL	0
6	visit_duration	INTEGER	1	'0'	0
7	incremented_omnibox_typed_score	BOOLEAN	1	FALSE	0
8	opener_visit	INTEGER	0	NULL	0

```
sqlite> PRAGMA table_info(downloads);
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	guid	VARCHAR	1	NULL	0
2	current_path	LONGVARCHAR	1	NULL	0
3	target_path	LONGVARCHAR	1	NULL	0
4	start_time	INTEGER	1	NULL	0
5	received_bytes	INTEGER	1	NULL	0
6	total_bytes	INTEGER	1	NULL	0
7	state	INTEGER	1	NULL	0
8	danger_type	INTEGER	1	NULL	0
9	interrupt_reason	INTEGER	1	NULL	0
10	hash	BLOB	1	NULL	0
11	end_time	INTEGER	1	NULL	0
12	opened	INTEGER	1	NULL	0
13	last_access_time	INTEGER	1	NULL	0
14	transient	INTEGER	1	NULL	0
15	referrer	VARCHAR	1	NULL	0
16	site_url	VARCHAR	1	NULL	0
17	tab_url	VARCHAR	1	NULL	0
18	tab_referrer_url	VARCHAR	1	NULL	0
19	http_method	VARCHAR	1	NULL	0
20	by_ext_id	VARCHAR	1	NULL	0
21	by_ext_name	VARCHAR	1	NULL	0
22	etag	VARCHAR	1	NULL	0
23	last_modified	VARCHAR	1	NULL	0
24	mime_type	VARCHAR(255)	1	NULL	0
25	original_mime_type	VARCHAR(255)	1	NULL	0
26	embedder_download_data	VARCHAR	1	`''''''`	0

Phase 3: Web Navigation & Transition Bitmask Decoding

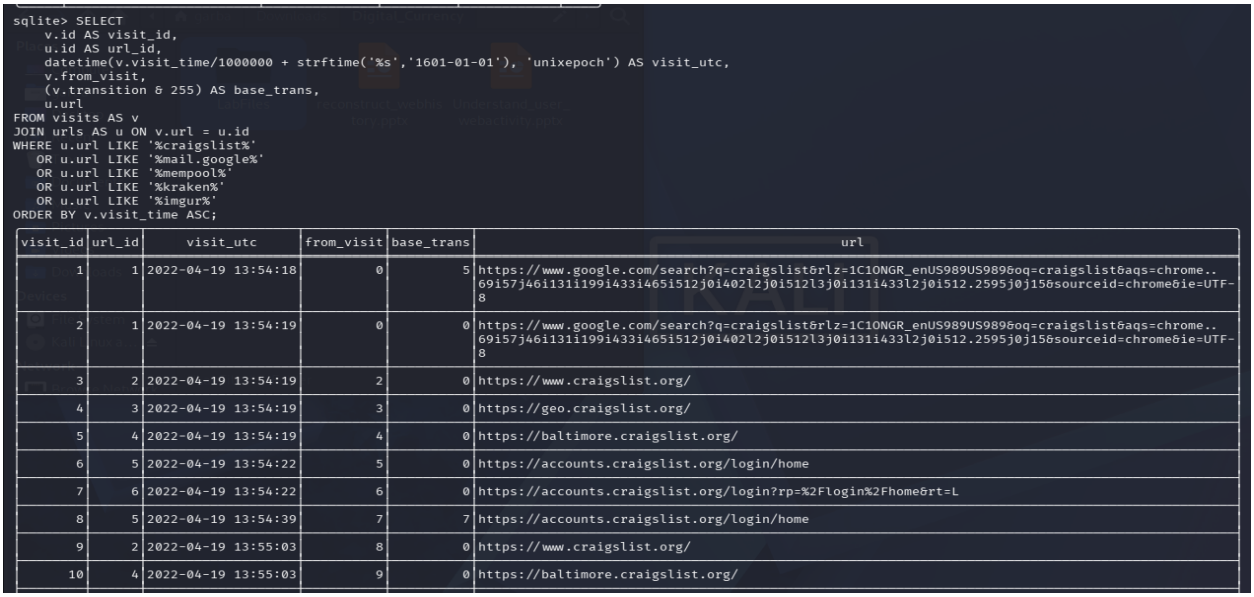
To reconstruct the user's browsing sequence, visits and urls tables were joined. WebKit microsecond timestamps were converted to UTC (datetime(visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch')). Base transition values were computed using bitwise logic (v.transition & 255) to distinguish user intent:

- **5 (GENERATED / OMNIBOX):** Address or query manually typed into the Omnibox.
- **0 (LINK):** Direct click on a hyperlink from a parent page (from_visit).
- **7 (FORM_SUBMIT):** Navigation resulting from submitting an HTML form.

SQL

```
SELECT v.id AS visit_id, u.id AS url_id,
    datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
    v.from_visit, (v.transition & 255) AS base_trans, u.url
FROM visits AS v JOIN urls AS u ON v.url = u.id
WHERE u.url LIKE '%craigslist%' OR u.url LIKE '%mail.google%'
  OR u.url LIKE '%mempool%' OR u.url LIKE '%kraken%' OR u.url LIKE '%imgur%'
ORDER BY v.visit_time ASC;
```

Figure A3: Terminal query output displaying chronological navigation history across Craigslist, including Google searches, page visits, and form submissions.



```
sqlite> SELECT
    v.id AS visit_id,
    u.id AS url_id,
    datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
    v.from_visit,
    (v.transition & 255) AS base_trans,
    u.url
FROM visits AS v
JOIN urls AS u ON v.url = u.id
WHERE u.url LIKE '%craigslist%'
   OR u.url LIKE '%mail.google%'
   OR u.url LIKE '%mempool%'
   OR u.url LIKE '%kraken%'
   OR u.url LIKE '%imgur%'
ORDER BY v.visit_time ASC;
```

visit_id	url_id	visit_utc	from_visit	base_trans	url
1	1	2022-04-19 13:54:18	0	5	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8
2	1	2022-04-19 13:54:19	0	0	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8
3	2	2022-04-19 13:54:19	2	0	https://www.craigslist.org/
4	3	2022-04-19 13:54:19	3	0	https://geo.craigslist.org/
5	4	2022-04-19 13:54:19	4	0	https://baltimore.craigslist.org/
6	5	2022-04-19 13:54:22	5	0	https://accounts.craigslist.org/login/home
7	6	2022-04-19 13:54:22	6	0	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
8	5	2022-04-19 13:54:39	7	7	https://accounts.craigslist.org/login/home
9	2	2022-04-19 13:55:03	8	0	https://www.craigslist.org/
10	4	2022-04-19 13:55:03	9	0	https://baltimore.craigslist.org/

Phase 4: Cryptocurrency, Communication & Download Artifact Extraction

Query execution against visits, urls, and downloads uncovered cryptocurrency verification, email communications, and local file retention:

1. **Kraken Exchange Activity:** Access to kraken.com/u/funding (15:01:21 UTC) and kraken.com/u/history/ledger (15:02:05 UTC), showing ledger views of **$41,563.70 USD**.
2. **Blockchain Explorer:** Direct lookup on Mempool.space at 15:01:38 UTC inspecting Bitcoin TXID 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 linked to address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt.
3. **Gmail Inquiries:** Navigation to Gmail inbox threads (unsub.fscs@gmail.com) between 15:02:32 UTC and 15:02:56 UTC discussing the transaction and listing.
4. **Local Download Record:** Querying downloads returned record ID #1, proving file proof_of_payment.png (33,844 bytes) was downloaded at 14:56:58 UTC and stored at C:\Users\FSCS_User\Desktop\proof_of_payment.png.

SQL

```
SELECT id AS download_id,
    datetime(start_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS start_utc,
    target_path, received_bytes
```

FROM downloads;

Figure A4: Terminal query output rendering Kraken ledger visits, Mempool blockchain lookups, Gmail threads, and the local download artifact record for proof_of_payment.png.

```
sqlite> SELECT
    id AS download_id,
    datetime(start_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS start_utc,
    target_path,
    received_bytes
FROM downloads;
```

download_id	start_utc	target_path	received_bytes
1	2022-04-19 14:56:58	C:\Users\FSCS_User\Desktop\proof_of_payment.png	33844

3. REQUIRED INVESTIGATION FINDINGS WORKSHEET

The following structured findings worksheet directly addresses the eight investigation tasks mandated by the SBT-DF204 assessment brief:

NO	INVESTIGATION TASK	VERIFIED FORENSIC FINDING & TECHNICAL CITATION
Q1	**Acquired Evidence File & Preservation**	Acquired History (b991b0fa69062c44caf599c468c8088576ebd9cebbf46d02c89cf51bcde3fd19). Set to read-only (chmod 444); created bit-stream duplicate History_working.sqlite for execution in sqlite3 v3.53.4 (Figure A1).
Q2	**Relevant History Tables & Fields**	Analyzed urls (id, url, title), visits (visit_time, from_visit, transition), downloads (target_path, received_bytes), and downloads_url_chains (Figure A2).
Q3	**Timeframe & Timestamp Conversion**	Converted WebKit epoch timestamps to UTC using datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch'). Activity occurred **19 April 2022** between **13:54:18 UTC and 15:02:56 UTC** (Figures A3, A4).
Q4	**Navigation & Transition Mechanics**	Evaluated (transition & 255) base values and from_visit links. Identified Omnibox search (base_trans = 5), hyperlink clicks

		(base_trans = 0), and login form submission (base_trans = 7) targeting accounts.craigslist.org/login/home (Figure A3).
Q5	**Posting & Sales Activity Evidence**	Extracted visit records to baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html and management portal post.craigslist.org/manage/7473121658 titled *"cheaper than Rx supplements"* (Figure A3).
Q6	**Communications & Payment Artifacts**	Isolated Gmail inbox unsub.fscs@gmail.com, Mempool Bitcoin lookup (517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5), Kraken ledger ($41,563.70), and downloaded file C:\Users\FSCS_User\Desktop\proof_of_payment.png (Figure A4).
Q7	**Chronological Activity Timing**	Complete timeline reconstructed on **Tuesday, 19 April 2022**, spanning classified posting, local file download, blockchain verification, exchange ledger review, and email confirmation (Figures A3, A4).
Q8	**Defensible Attribution & Conclusion**	Browser profile FSCS_User executed trafficking workflow. Confidence: **Moderate-to-High (Profile Level)** / **Low-to-Moderate (Human Level)**. Limitations: Lack of physical keyboard logs, shared endpoint access, and uncaptured HTML body text.

4. CHRONOLOGICAL FORENSIC TIMELINE

The following timeline details the exact chronological sequence of material browser events reconstructed from History_working.sqlite. All timestamps reflect UTC.

Timestamp (UTC)	Record ID	Artifact / URL	Base Trans	Forensic Significance & Observed Event
2022-04-19 13:54:18	Visit #1	google.com/search?q=craigslist...	5 (Omnibox)	Initial search query for Craigslist executed in address bar.
2022-04-19 13:54:19	Visit #5	https://baltimore.craigslist.org/	0 (Link)	Navigated to regional Baltimore Craigslist board via search result.
2022-04-19 13:54:39	Visit #8	accounts.craigslist.org/login/home	7 (Form)	Form submission event executing account login for post management.
2022-04-19 13:55:03	Visit #10	post.craigslist.org/manage/7473121658	0 (Link)	Managing active listing *"cheaper than Rx supplements"* (7473121658).
2022-04-19 14:56:58	DL #1	Desktop\proof_of_payment.png	N/A	Downloaded and saved proof image (33,844 bytes) to desktop.
2022-04-19 15:01:21	Visit #66	https://www.kraken.com/u/funding	0 (Link)	Inspected Kraken exchange funding balance page ($41,543.80).

2022-04-19 15:01:38	Visit #70	mempool.space/tx/517b21569149...	0 (Link)	Looked up Bitcoin TXID 517b2156914944339a96137ad8978408ea52b2fc....
2022-04-19 15:02:05	Visit #71	https://www.kraken.com/u/history/ledger	0 (Link)	Inspected Kraken ledger account history ($41,563.70 USD).
2022-04-19 15:02:32	Visit #72	mail.google.com/mail/u/0/#inbox/...	48 (Redirect)	Accessing Gmail thread (unsub.fscs@gmail.com) containing TXID.

5. ATTRIBUTION ASSESSMENT & TECHNICAL LIMITATIONS

Observed Facts vs. Technical Inferences

- **Direct Forensic Observations:** SQLite database records prove that browser profile FSCS_User executed searches, managed Craigslist post 7473121658, downloaded proof_of_payment.png, checked Bitcoin transaction 517b2156... on Mempool, reviewed Kraken exchange ledgers ($41,563.70), and accessed Gmail account unsub.fscs@gmail.com.
- **Reasonable Inferences:** It is inferred that the primary user of the device operated the browser profile to conduct, verify, and track payment for illicit supplement sales.

Technical Limitations & Alternative Defense Hypotheses

1. **User Profile vs. Physical Identity:** Chrome SQLite records track activity within a user profile (FSCS_User). History artifacts cannot prove who was physically pressing the keys without secondary endpoint logs (e.g., OS logon events, CCTV, or biometric capture).
2. **Shared Terminal / Unlocked Endpoint:** An unauthorized individual could have accessed an unattended, unlocked laptop operating under the logged-in profile.
3. **Uncaptured Web Page Bodies:** History databases record URLs, titles, and timestamps, but do not retain un-cached rendered HTML body text or private email message bodies.

6. EVIDENCE LOG APPENDIX

Item ID	Source Record / Artifact	Forensic Finding & Technical Observation	Screenshot Ref
E01	File System / sha256sum	b991b0fa69062c44caf599c468c8088576ebd9cebbf46d02c89cf51bcde3fd19	Figure A1 (Screenshot 479 & Screenshot 480)
E02	PRAGMA table_info	Verified column structure for urls, visits, and downloads	Figure A2 (Screenshot 481 & Screenshot 482)
E03	visits JOIN urls	Omnibox search (base_trans=5) and Craigslist management access (7473121658)	Figure A3 (Screenshot 483)
E04	downloads Table	Local download record: 14:56:58 UTC, Desktop\proof_of_payment.png (33,844 bytes)	Figure A4 (Screenshot 484)
E05	visits JOIN urls	Mempool lookup of TXID 517b2156914944339a96137ad8978408ea52b2fc... at 15:01:38 UTC	Figure A4 (Screenshot 484)
E06	visits JOIN urls	Kraken Ledger view ($41,563.70) at 15:02:05 UTC and Gmail thread access	Figure A4 (Screenshot 484)

7. ACADEMIC INTEGRITY & AUTHORIZATION STATEMENT

I confirm that I completed these digital forensics case study independently in an authorized virtual environment using the supplied historical Chrome SQLite scenario dataset. I have accurately documented my own SQL queries, record citations, evidence log entries, cryptographic hashes, and forensic observations without fabrication.

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Student ID: 2025/FWSD/11463

Course Code: SBT-DF204

Date: October 11, 2026